

Answer **all** the questions in the spaces provided.

- 1 A projectile is fired from ground level with initial velocity u at an angle θ to the horizontal as shown in Fig. 1.1. The projectile strikes a target which is at a horizontal displacement x from the point of projection and a vertical height y above ground level.

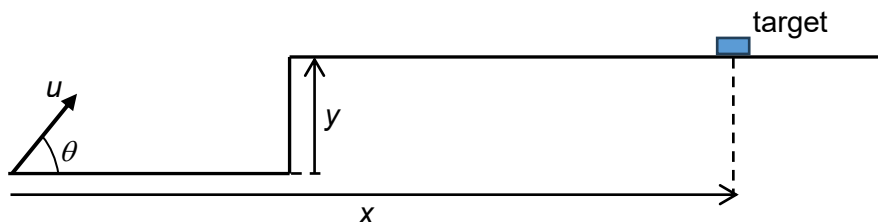


Fig. 1.1

- (a) Neglecting the effect of air resistance, show that the vertical height y is given by the expression

$$y = x \tan \theta - 4.91 \left(\frac{x}{u \cos \theta} \right)^2$$

$$s_x = u_x t$$

$$x = (u \cos \theta) t$$

$$t = \frac{x}{u \cos \theta} \quad [\text{M1}]$$

$$s_y = u_y t + \frac{1}{2} a_y t^2$$

$$y = (u \sin \theta) t + \frac{1}{2} (-9.81) t^2 \quad [\text{C1}]$$

$$y = u \sin \theta \times \frac{x}{u \cos \theta} - 4.91 \left(\frac{x}{u \cos \theta} \right)^2 \quad [\text{M1}]$$

$$y = x \tan \theta - 4.91 \left(\frac{x}{u \cos \theta} \right)^2 \quad [\text{A0}]$$

[3]

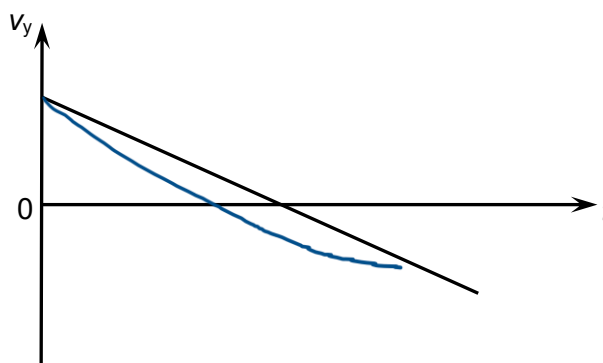
- (b) Given that the angle θ is 60° , the horizontal displacement x is 115 m and the vertical height y is 23 m, calculate the speed u .

$$23 = 115 \tan 60 - 4.91 \left(\frac{115}{u \cos 60} \right)^2$$

$$u = 38 \text{ m s}^{-1}$$

$$u = \dots\dots\dots \text{ m s}^{-1} \quad [1]$$

- (c) Fig. 1.2 shows the variation with time t of the vertical velocity v_y of the projectile when air resistance is negligible. On the same axes, sketch a graph to show the variation with time t of the vertical velocity v_y of the projectile when air resistance is not negligible. [2]



The curve must decrease at a decreasing rate throughout, with following details:

1. Positive area larger than negative area
2. a larger initial gradient
3. a smaller t -intercept
4. at $v_y = 0$, gradient parallel

Fig. 1.2

[Total: 6]

